Block Chain

Technical design overview

1. The Chain: Time, Blocks, and Rounds

The Block Chain organises activity into a fixed heartbeat. Every **5 seconds** one "tick" (called a **slot**) passes. During each tick, one elected participant may add a bundle of activity (a **block**) to the chain. A fixed number of ticks (currently about **7 days** of slots) forms a **round** (called an **epoch**) at the end of which participants for the next round are elected.

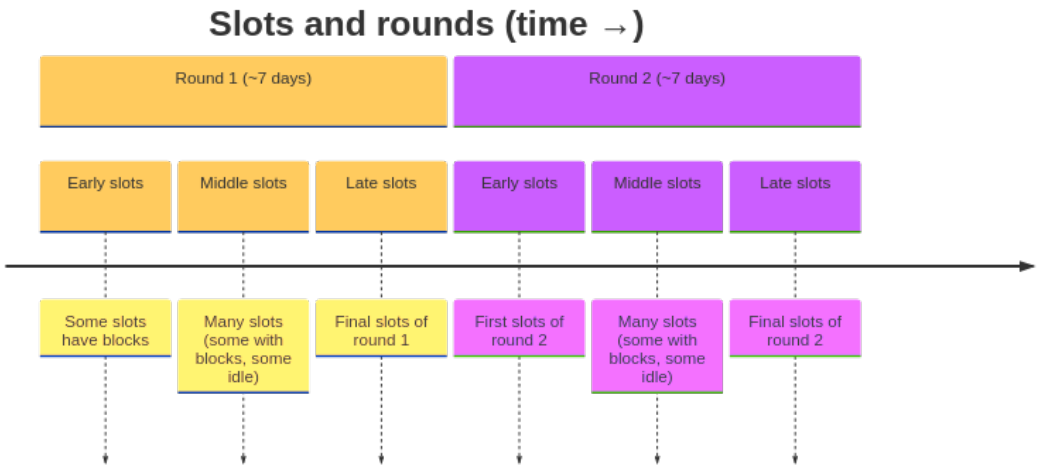

Figure 1: Slots and rounds (time)

Concept	Plain meaning
Tick (slot)	5-second window. At most one block is added per tick.
Round (epoch)	7 days. Block producers for the next round are chosen at the end of each round, weighted by their stake.

Concept	Plain meaning
Block	A sealed package of activity: transfers, account changes, and other records — linked to the block before it.

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2. What Is Inside a Block

Each block is a tamper-proof envelope. Once added to the chain it cannot be changed.

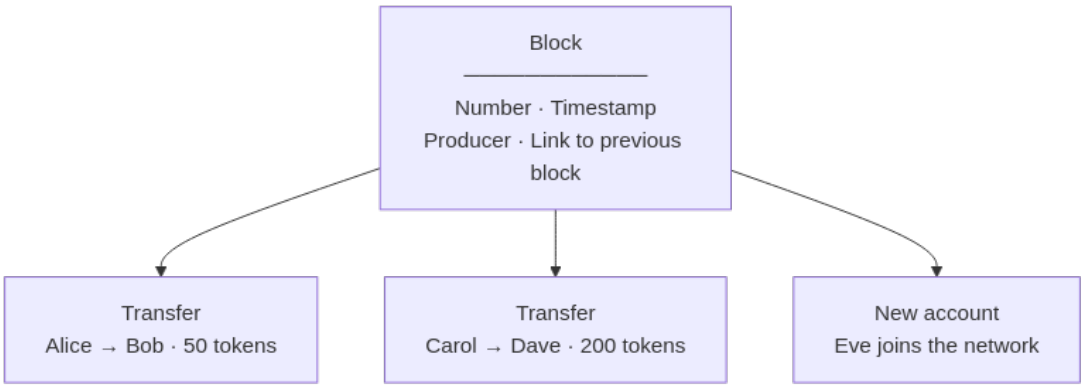

Figure 2: Block contents

Types of activity a block can record

Activity	What it does
Launch	Sets up the network for the first time
New account	Registers and funds a new participant
Transfer	Moves tokens between accounts
Update account	Participant updates their own profile
Renew account	Network keeps an account active
Close account	Network closes an account that can no longer pay its upkeep

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3. Network Participants: Beacon, Relays, and Miners

Three types of node keep the network running. They have distinct, non-overlapping roles so that no single participant can act alone to alter the chain. In the rare event that the original Beacon is permanently lost, a Relay that already holds the full chain can be promoted to become the new Beacon, preserving continuity.

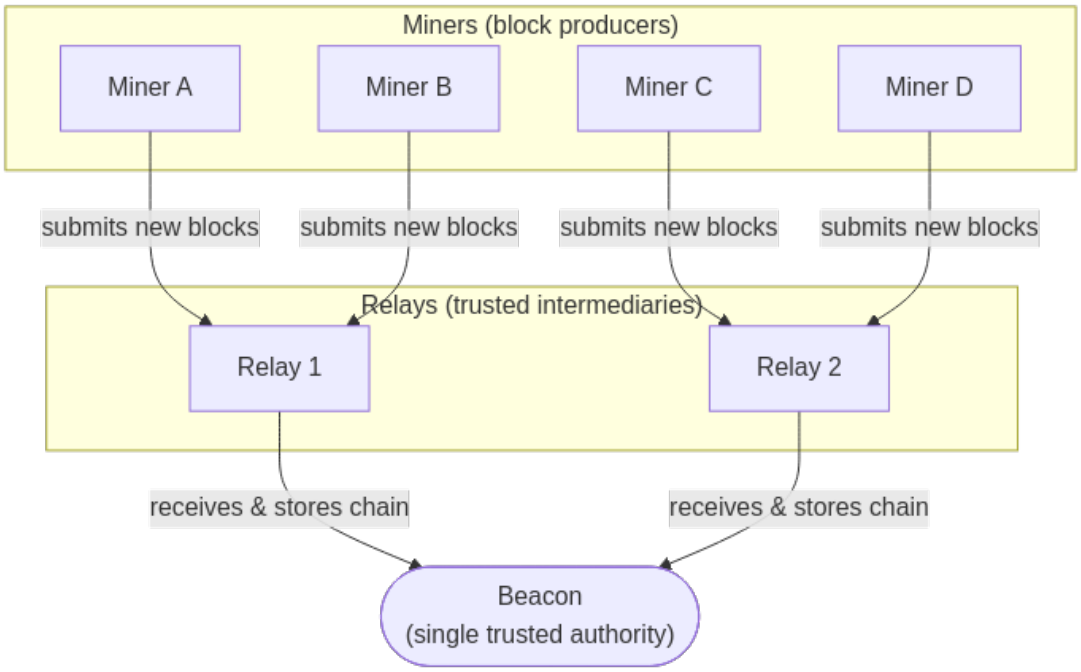

Figure 3: Beacon, Relays, and Miners

Role	What they do	Adds blocks?	Sees full history?
Beacon	Single authoritative record-keeper; validates everything	No	Yes
Relay	Trusted gateway — distributes the chain to miners, shields the Beacon	No	Yes
Miner	Elected participant who packages and adds new blocks; earns fees	Yes	partial

Why this matters: Miners compete fairly — only the randomly

elected miner for a given tick may add a block, and their election odds are proportional to how much they have staked. This makes the network both decentralised and predictable.

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4. Checkpoints — Keeping the Network Lean

As the chain grows over months and years, storing every block from the very beginning becomes expensive. **Checkpoints** solve this: starting from launch day, the network marks confirmed ranges of blocks whose combined contents fully determine the current state of every account. The first checkpoint is the network launch itself; later checkpoints cover ranges between two neighbouring marks. New participants can join from a checkpoint and only need to read the blocks between that checkpoint and the previous one, instead of replaying the entire history.

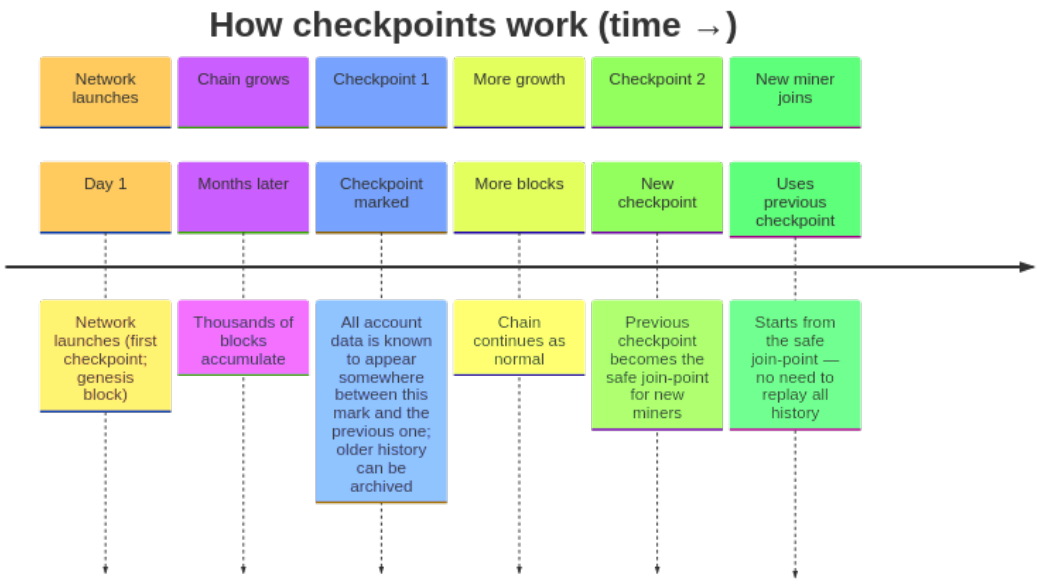

Figure 4: How checkpoints work (time)

Term	Meaning
Snapshot (checkpoint)	A marker in the chain that guarantees every account's latest state can be reconstructed from the blocks between this mark and the previous one (states are not required to sit in a single block)
Safe join-point	The checkpoint before the latest one — proven stable and widely agreed upon

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5. Reserved Accounts — Issuing Tokens

The network reserves a range of special accounts for issuing and managing tokens. These accounts are allowed to show negative balances as an accounting device (similar to a central bank's balance sheet) and can create new token types.

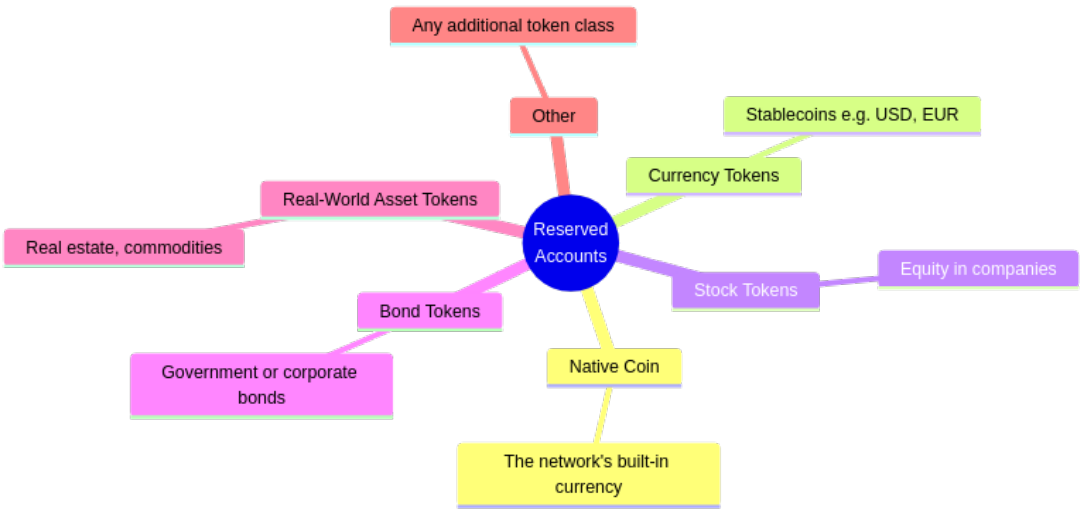

Figure 5: Reserved Accounts

Account	Purpose
Genesis (account 0)	Issues the network's native coin; sets initial supply

Account	Purpose
Fee collector (account 1)	Receives small upkeep fees paid by all accounts
Reserve (account 2)	Holds unallocated supply
Recycle (account 3)	Receives balances from closed accounts
Accounts 4 and above	Available for currency, stock, bond, RWA, and other token classes

Each token type is issued from its own dedicated reserved account, giving issuance a clear, auditable home on-chain.

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6. User Accounts — Holding and Doing

Every person or organisation on the network has a **user account**. An account can hold any mix of tokens (native coin, stablecoins, equity, bonds, etc.) and carry a personal data attachment that can grow to include digital collectibles and self-executing contracts.

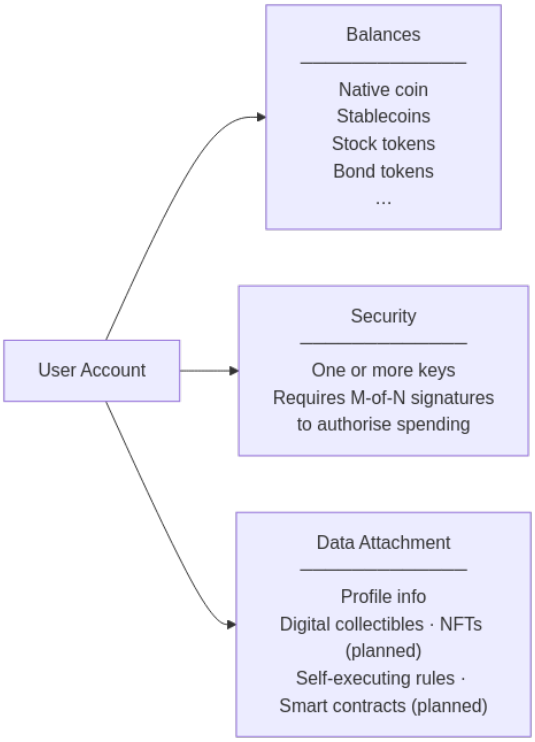

Figure 6: User Account

Capability	Available today
Hold multiple token types	Yes
Require more than one key to spend (multi-sig)	Yes
Attach profile or custom data	Yes
Own digital collectibles (NFTs)	Planned
Self-executing rules (smart contracts)	Planned

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7. What Is Built and What Is Next

Core Chain

Capability	Status
Predictable, time-based block production	Live
Stake-weighted, tamper-proof leader election	Live
Immutable transaction records	Live

Capability	Status
Multi-token balances	Live
Small, usage-based fees	Live
Periodic checkpoints to keep storage lean	Live

Network

Capability	Status
Beacon — authoritative record-keeper	Live
Relays — scalable miner gateway	Live
Miners — decentralised block production	Live
Miner fast-join via snapshots	Live

Security note: The current network stack focuses on correctness and basic robustness. Advanced protections against large-scale abusive behaviour (such as massive connection floods or automated probing) are not yet implemented and should be assumed to require additional hardening before internet-wide deployment.

Tokens & Accounts

Capability	Status
Native coin	Live
Custom token issuance (by reserved accounts)	Live
User account registration	Live
Account upkeep and closure	Live
Digital collectibles (NFTs)	Planned
Self-executing rules (smart contracts)	Planned
Dedicated currency / stock / bond / RWA token classes	Planned

Interfaces

Capability	Status
Command-line tool	Live
REST (HTTP) API	Live
JavaScript / Node.js library	Live
MCP (Model Context Protocol) server	Live
Real-time streaming API	Planned
Web explorer / dashboard	Planned

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8. Key Differences vs Other Chain Designs

This section summarises how our design differs from many other blockchain systems in two areas: consensus/network and accounts/checkpoints.

8.1 Consensus and Network

Aspect	our design	Typical other chains
Authority model	Single **Beacon** as authoritative record-keeper; **Relays** as trusted gateways; **Miners** only produce blocks. No single role can alter the chain alone.	Often all full nodes are equal (e.g. Bitcoin) or validators both propose and validate (e.g. many PoS chains).
Block production	Only **Miners** add blocks; Beacon and Relays never produce blocks. Miners are elected per slot and connect via Relays (or optionally to Beacon).	Validators or miners usually talk to each other in a flat or mesh topology; no dedicated “authority + gateway” split.

Aspect	our design	Typical other chains
Visibility	Miners see only a **partial** history (needed for proposing blocks); full chain lives at Beacon and Relays.	Full nodes and validators typically store and validate the full chain.
Recovery	A Relay that holds the full chain can be **promoted to Beacon** if the original Beacon is lost, preserving continuity without changing the protocol.	Failover is usually handled by out-of-band replacement or social consensus, not a defined “next Beacon” role.
Time and slots	Fixed **5-second slots** and at most one block per slot; stake-weighted, VRF-based leader election (Ouroboros-style).	Variable block times or multiple blocks per “round” are common; leader selection varies by chain.

In short: our design separates **who keeps the truth** (Beacon), **who distributes it** (Relays), and **who extends it** (Miners), instead of merging these into one validator set.

8.2 Account and Checkpoint

Aspect	our design	Typical other chains
Checkpoints / sync	**Checkpoints** mark confirmed ranges; new participants join from a checkpoint and only process blocks **between that checkpoint and the previous one**. No need to replay from genesis.	New nodes often sync from genesis or from a recent “snapshot” that may be a full state dump rather than a chain-range guarantee.
State and history	Checkpoints guarantee that account state can be **reconstructed from the blocks** in the range between two consecutive checkpoints; state is not required to live in a single block.	State is usually derived from applying all blocks (or from a state DB); “checkpoint = chain range” is not always explicit.
Reserved accounts	**Fixed reserved accounts** (Genesis, Fee collector, Reserve, Recycle, then token issuers) with defined roles; some may have negative balances as an accounting device.	Often a single “system” or treasury address, or no formal reserved range; token issuance is contract- or protocol-specific.

Aspect	our design	Typical other chains
Account lifecycle	Explicit **registration, upkeep (fees), renewal, and closure**; closed-account balances go to Recycle. Accounts can be closed by the network when they cannot pay upkeep.	Accounts are often “create once, use forever” or closed only by user action; no built-in upkeep or network-initiated closure.
Multi-token and issuance	Native **multi-token** balances per account; each token type is issued from a **dedicated reserved account** (4+), giving issuance a clear on-chain home.	Multi-token is usually via contracts or side tables; issuance is not always tied to a fixed reserved-account model.

In short: our design treats **checkpoints as bounded chain ranges** that define state reconstruction and lean sync, and **accounts** as first-class objects with lifecycle, upkeep, and reserved-account-based token issuance, rather than only “address + balance” or contract-only issuance.